

# Inside the internet

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.3 Computer networks, connections and protocols — Paper 1, Computer systems                             |
| Lesson          | Inside the internet (lesson 5 of the unit)                                                                        |
| Lesson ID       | nw-105                                                                                                            |
| Experience ID   | nw-105                                                                                                            |
| Plan ID         | PLAN-NW-L05                                                                                                       |
| Content version | 8723d5136ac2                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=nw-105">https://revise360.co.uk/experience.html?id=nw-105</a> |

## Big question, outcomes and success criteria

**Big question** — How does the internet work?

### Prerequisites

The lesson before this one is nw-104. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: nw-101, nw-102, nw-103.

### Assessable outcomes

1. **nw-105-o1** — understand what the internet actually is
2. **nw-105-o2** — understand the terms Domain Name Service and domain name server
3. **nw-105-o3** — understand what is meant by hosting and the cloud
4. **nw-105-o4** — understand what is meant by the terms web server and client

## Vocabulary, resources and preparation

**Key vocabulary** — DNS (Domain Name Service), Hosting, The cloud

### Resources

|                   |                                                    |
|-------------------|----------------------------------------------------|
| SLIDES            | presentations/NW_L05_InsideTheInternet_Lesson.pptx |
| WORKSHEET         | worksheets/NW_L05_InsideTheInternet_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=nw-105                          |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on <code>NW_LO5_InsideTheInternet_Worksheet.docx</code> is filled in.  |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find What is the internet?.                           | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. What is the internet?

- A worldwide collection of computer networks: the biggest WAN.
- Networks are joined by routers and linked by cables, including undersea fibre.
- The World Wide Web is a service that uses the internet; it isn't the internet itself.

**Challenge on this wall:** Define the internet in one sentence.

### 2. Domain Name Service (DNS)

- People use domain names like www.google.com; computers use IP addresses.
- DNS turns a domain name into the matching IP address.
- DNS is made up of many domain name servers, not just one.

**Challenge on this wall:** Why don't we just type IP addresses into our browsers?

### 3. Hosting

- Hosting means storing websites, files or services on a server so others can reach them over the internet.
- Hosting companies run data centres full of servers and rent out space.

**Challenge on this wall:** Why would a small business pay a hosting company instead of running its own web server?

### 4. The cloud

- Storage, software and services on remote servers, accessed over the internet.
- Examples: Google Drive and Microsoft OneDrive.
- You can reach your files from any device, anywhere with a connection.

**Challenge on this wall:** What is the advantage of working in the cloud, rather than installing programs and saving files locally?

### 5. Web servers and clients

- A web server stores web pages and sends them to clients when they are requested.
- A client, such as the browser on your phone, requests a page and displays it.

**Challenge on this wall:** Describe the client-server relationship between a web server and a mobile phone.

### 6. File servers

- A file server stores files centrally for everyone on a network.
- A desktop PC is a client: it requests a file, and the file server sends it.

**Challenge on this wall:** Describe the relationship between a file server and a desktop PC.

## Retrieval and the check for understanding

**Starter** — You save a file on a school computer and open it later at home. Describe what has to happen for that to work.

On the board or the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (nw-105-s4-c1), so the wording a pupil sees matches the wording you say.

**Which is the best description of the cloud?**

**Correct:** Storage and services on remote servers, reached over the internet

### What a pupil might choose instead

- Wireless networks in the sky
- Files saved on a USB stick
- A type of weather satellite

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** 'The cloud' is just other people's servers, in data centres, that you use over the internet.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

| STAGE     | ACTIVITIES         | WHAT IT IS                                                                                                                |
|-----------|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | <a href="#">11</a> | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | <a href="#">2</a>  | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #                     | STATION                           | ACTIVITIES        | RECORD ON PAPER                       |
|-----------------------|-----------------------------------|-------------------|---------------------------------------|
| <a href="#">1</a>     | <b>What is the internet?</b>      | <a href="#">2</a> | The internet is...                    |
| <a href="#">2</a>     | <b>Domain Name Service (DNS)</b>  | <a href="#">2</a> | DNS is used to...                     |
| <a href="#">3</a>     | <b>Hosting</b>                    | <a href="#">2</a> | Hosting is...                         |
| <a href="#">4</a>     | <b>The cloud</b>                  | <a href="#">2</a> | The cloud means...                    |
| <a href="#">5</a>     | <b>Web servers and clients</b>    | <a href="#">2</a> | A web server... The client...         |
| <a href="#">6</a>     | <b>File servers</b>               | <a href="#">2</a> | A file server...                      |
| <a href="#">final</a> | <b>Final challenge: the cloud</b> | <a href="#">1</a> | the answer to the station's challenge |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

## Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                                                   | MARKS | WHERE                                                  |
|-----------|--------------------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| nw-105-x1 | Describe what the internet is.                                                             | 2     | worksheet, Exam practice a); deck, Exam practice slide |
| nw-105-x2 | Explain the steps DNS goes through when a browser looks up a web address.                  | 4     | worksheet, Exam practice b); deck, Exam practice slide |
| nw-105-x3 | A school is considering moving its files to the cloud. Discuss the benefits and the risks. | 8     | worksheet, Exam practice c); deck, Exam practice slide |

## Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/nw-105.json`, which is not published with the website.

## Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is the internet?** — All the websites in the world; One huge computer owned by a tech company
- **What is the World Wide Web?** — Another name for the internet; The cables that link countries
- **What does DNS do?** — Stores every web page; Gives devices their MAC addresses
- **What is meant by hosting?** — Inviting people to use your Wi-Fi; Buying a domain name
- **Why might a business use a hosting company?** — Its website becomes invisible to competitors; It no longer needs a domain name

## Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- What the internet is
- How DNS works
- Hosting, the cloud, web servers and clients

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is nw-106, Star and mesh networks. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

---

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

---

Revise 360 — a product of Revise 360 Ltd, created by Olly Pettitt. An independent resource: not written, endorsed or approved by OCR or any exam board. Classroom use and personal study only.